

Erratum and Addendum: Supplementary §S2 and the deferred proofs of Theorems 2, 3, Corollary 4, Proposition 5, and Theorem 7 to “The Price of Meaning: Why Every Semantic Memory System Forgets”

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Notice

The arXiv preprint *The Price of Meaning: Why Every Semantic Memory System Forgets* (arXiv:2603.27116v1) states, immediately after the proof sketch of Theorem 1 (Semantic Spectral Bound):

“Full proof in Supplementary §S2.”

(main paper, after Theorem 1)

The Supplementary §S2 referenced in that statement was not included in the original submission. In addition, the proofs of Theorems 2, 3, Corollary 4, Proposition 5, and Theorem 7 were stated in the main text without accompanying derivations. This erratum supplies all of these omitted derivations in full. Nothing in the main paper changes: no theorem statement, definition, axiom, figure, table, or numerical result is affected. The proof sketch of Theorem 1 in the main paper is unchanged and remains correct; the material below expands it line by line, and supplies the proofs that were not written out for Theorems 2, 3, Corollary 4, Proposition 5, and Theorem 7. Theorem 6 already carried its (one-line) proof in the main paper and is not repeated here.

The document is organised as follows. §S1 collects the notation and preliminaries used throughout. §S2 gives the complete proof of Theorem 1 in four steps. §S3 proves Theorem 2 (Positive Cap Mass). §S4 proves Theorem 3 (Inevitable Forgetting), Corollary 4 (Stretched Exponential), and Proposition 5 (Population Power Law). §S5 derives Theorem 7 (No Escape) from the preceding results. References cited in this erratum are listed at the end. Cross-references to numbered objects in the main paper (axioms, theorems, equations, figures) carry their original numbering.

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S1 Notation and preliminaries

Items live in a measurable space $(\mathcal{X}, \mathcal{F})$ with a Borel probability measure P_X . The semantic kernel $K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ of Axiom A2 is symmetric, continuous, and positive semidefinite. By Mercer’s theorem applied to K on $L^2(\mathcal{X}, P_X)$, there is a non-increasing sequence of non-negative eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$ with $\sum_j \lambda_j < \infty$ and an orthonormal system of eigenfunctions $\{\psi_j\}_{j \geq 1}$ in $L^2(P_X)$ such that

$$K(x, x') = \sum_{j \geq 1} \lambda_j \psi_j(x) \psi_j(x'),$$

with convergence in $L^2(P_X \otimes P_X)$ and uniform on compact sets when K is continuous [1]. The associated reproducing kernel Hilbert space (RKHS) is $\mathcal{H}_K$, and the canonical feature map is $\phi(x) = (\sqrt{\lambda_j} \psi_j(x))_{j \geq 1}$ with $\langle \phi(x), \phi(x') \rangle_{\mathcal{H}_K} = K(x, x')$.

For a measurable encoder $T : \mathcal{X} \rightarrow \mathcal{Z}$ taking values in a Polish space $\mathcal{Z}$, write $\sigma(T)$ for the sub- σ -algebra of $\mathcal{F}$ generated by T . Let $Y \in \{0, 1\}$ denote the (latent) retrieval relevance label for a query q , and let $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$. Throughout we assume η is $\sigma(\phi)$ -measurable; this is exactly Axiom A2 of the main paper.

The Bayes retrieval risk under a $\{0, 1\}$ -loss is

$$\mathcal{R}^*(T) = \inf_{h \in \mathcal{M}(\sigma(T))} \mathbb{P}[h(T(X)) \neq Y],$$

where $\mathcal{M}(\sigma(T))$ denotes measurable functions of $T(X)$. The distortion budget refers to the rate-distortion functional $D(T) = \mathbb{E}[d(\phi(X), \Phi_T)]$, where Φ_T is the optimal reconstruction of $\phi(X)$ given $T(X)$ in squared error and d is the squared Hilbert-space distance.

Given the spectral decomposition, define for each threshold $\gamma > 0$ the truncated semantic statistic

$$\Phi_\gamma(x) = (\sqrt{\lambda_j} \psi_j(x))_{\lambda_j > \gamma}, \quad r(\gamma) = \#\{j : \lambda_j > \gamma\}.$$

For Axiom A4 we use the pushforward $\mu = \phi_\# P_X$ on the semantic manifold, with intrinsic dimension $d_{\text{loc}}(z)$ at μ -almost every anchor z . A retrieval cap at anchor z with radius θ is

$$C(z, \theta) = \{x \in \mathcal{X} : \langle \phi(z), \phi(x) \rangle_{\mathcal{H}_K} \geq \cos \theta\}.$$

Axiom A4 states $c_1 r^{d_{\text{loc}}(z)} \leq \mu(B(z, r)) \leq c_2 r^{d_{\text{loc}}(z)}$ for Hilbert-space balls $B(z, r)$ with $0 < r < r_0(z)$.

S2 Full proof of Theorem 1 (Semantic Spectral Bound)

We restate the theorem for the reader’s convenience.

Theorem (Semantic Spectral Bound; main paper, Theorem 1). *Let K be the semantic kernel of Axiom A2 with Mercer eigenpairs $(\lambda_j, \psi_j)_{j \geq 1}$. Under Axioms A1–A3, for every encoder T that is rate-distortion optimal at distortion level D there exists a threshold $\gamma(D) > 0$ such that:*

- (i) *T factors P_X -almost surely through the truncated semantic statistic $\Phi_{\gamma(D)}$;*
- (ii) *the semantically useful effective rank satisfies $d_{\text{eff}}(T) \leq r(\gamma(D)) = \#\{j : \lambda_j > \gamma(D)\}$;*
- (iii) *adding nominal dimensions to T without adding new modes with $\lambda_j > \gamma(D)$ does not reduce $\mathcal{R}^*(T)$ and does not change $d_{\text{eff}}(T)$.*

The argument has four steps: (A) reduction to a sufficient semantic statistic via Blackwell sufficiency under Axiom A2; (B) reduction to a kernel projection via Axiom A1; (C) reverse water-filling under the distortion budget given by Axiom A3, yielding the threshold $\gamma(D)$; (D) consequences for $d_{\text{eff}}(T)$ and for nominal-dimension padding.

Step A: Sufficiency of the semantic statistic. By Axiom A2, $\eta(x) = \mathbb{P}(Y = 1 \mid X = x)$ is $\sigma(\phi)$ -measurable. The Doob–Dynkin lemma [2] gives a Borel function $\tilde{\eta}$ such that $\eta(x) = \tilde{\eta}(\phi(x))$ for P_X -almost every x . Hence $\phi(X)$ is a sufficient statistic for Y in the sense of Blackwell [3, 4]: for any encoder T , the conditional distribution of Y given $T(X)$ is the Markov-kernel composition of the conditional distribution of Y given $\phi(X)$ with the conditional distribution of $\phi(X)$ given $T(X)$. Blackwell’s theorem then yields

$$\mathcal{R}^*(T) \geq \mathcal{R}^*(\phi),$$

with equality if and only if T contains a sufficient statistic for Y , equivalently if and only if $\eta(X) = \mathbb{E}[\eta(\phi(X)) \mid T(X)]$ holds P_X -almost surely. Any encoder achieving the Bayes risk therefore satisfies

$$\eta \text{ is } \sigma(T)\text{-measurable up to } P_X\text{-null sets.} \quad (\text{S.1})$$

Write $\mathcal{S}(T) = \sigma(T) \cap \sigma(\phi)$. From (S.1), η is $\mathcal{S}(T)$ -measurable. Any sub- σ -algebra of $\sigma(\phi)$ corresponds, by the spectral decomposition, to a measurable function of the Mercer coordinates $(\psi_j)_{j \geq 1}$. Define the *effective spectral support* of T as

$$S(T) = \{j \geq 1 : \psi_j \text{ is } \sigma(T)\text{-measurable up to } P_X\text{-null sets}\}.$$

Directions $j \notin S(T)$ are the *nuisance directions*: components of $\phi(X)$ along which T carries no information.

By Axiom A2, η depends on ϕ only through modes with $\lambda_j > 0$; modes with $\lambda_j = 0$ are irrelevant to $\mathcal{R}^*$. Combined with (S.1), any Bayes-optimal T must have $S(T) \supseteq \{j : \lambda_j > 0 \text{ and } \psi_j \not\perp \eta\}$.

Step B: Kernel-threshold scores factor through Hilbert-space projections. Axiom A1 specifies the retrieval score $s(q, x) = g(\langle w_q, \phi(x) \rangle_{\mathcal{H}_K})$ with g monotone increasing. Monotone transformations preserve risk minimisation, so the Bayes-optimal threshold rule is fully determined by the linear functional $f_q(x) = \langle w_q, \phi(x) \rangle_{\mathcal{H}_K}$. Expanding in the Mercer basis,

$$f_q(x) = \sum_{j \geq 1} w_{q,j} \sqrt{\lambda_j} \psi_j(x), \quad w_{q,j} = \langle w_q, \sqrt{\lambda_j} \psi_j \rangle.$$

Let $\Pi_J : \mathcal{H}_K \rightarrow \mathcal{H}_K$ denote orthogonal projection onto $\overline{\text{span}}\{\sqrt{\lambda_j} \psi_j : j \in J\}$ for any $J \subseteq \mathbb{N}$. The score induced by T is realised by the projected feature map $\Pi_{S(T)}\phi$, and the Bayes risk under T equals the Bayes risk of threshold rules on $\langle w_q, \Pi_{S(T)}\phi(\cdot) \rangle$. The retrieval risk of T depends on T only through $S(T)$.

Step C: Reverse water-filling under the distortion budget. Axiom A3 imposes a rate or distortion budget D . We work with distortion; the rate version is dual and yields the same threshold structure [5]. The minimum mean-squared reconstruction error of $\phi(X)$ given $T(X)$ decomposes along the Mercer basis as

$$D(T) = \sum_{j \notin S(T)} \lambda_j + \sum_{j \in S(T)} (\lambda_j - \text{Var}(\sqrt{\lambda_j} \psi_j(X) \mid T(X))).$$

For a fixed subset $S(T)$, the second sum is non-negative and is minimised at zero by perfectly preserving the retained coordinates. The distortion collapses to the tail mass

$$D(T) = \sum_{j \notin S(T)} \lambda_j.$$

Minimising tail mass subject to $D(T) \leq D$ is a reverse water-filling problem on the Mercer spectrum [5]: among encoders with retrieval risk at most that of $\Pi_S \phi$, the one with smallest tail mass retains modes in decreasing order of λ_j . Define

$$\gamma(D) = \inf \left\{ \gamma > 0 : \sum_{\lambda_j \leq \gamma} \lambda_j \leq D \right\}.$$

The rate-distortion-optimal retained set is $S^*(D) = \{j : \lambda_j > \gamma(D)\}$. Any other retained set S with the same Bayes risk satisfies $\sum_{j \notin S} \lambda_j \geq \sum_{j \notin S^*(D)} \lambda_j$, with equality if and only if $S = S^*(D)$ up to P_X -null sets. Hence every rate-distortion-optimal encoder satisfies

$$S(T) = \{j : \lambda_j > \gamma(D)\}, \quad T \text{ factors through } \Phi_{\gamma(D)}. \quad (\text{S.2})$$

This establishes claim (i).

Step D: Effective rank and nominal-dimension invariance. The semantically useful effective rank of T is the smallest dimension of a Euclidean representation that achieves $\mathcal{R}^*(T)$,

$$d_{\text{eff}}(T) = \inf \{k : \exists U : \mathcal{Z} \rightarrow \mathbb{R}^k, \mathcal{R}^*(U \circ T) = \mathcal{R}^*(T)\}.$$

By Step B, $\mathcal{R}^*(T)$ depends only on $S(T)$, and by (S.2) the optimal $S(T)$ is $\{j : \lambda_j > \gamma(D)\}$, which has cardinality $r(\gamma(D))$. Hence

$$d_{\text{eff}}(T) = r(\gamma(D)) = \#\{j : \lambda_j > \gamma(D)\},$$

establishing claim (ii).

For claim (iii), suppose T' is obtained from T by adjoining nominal coordinates $T_{\text{new}} : \mathcal{X} \rightarrow \mathbb{R}^m$ such that $\sigma(T_{\text{new}}) \subseteq \sigma(\phi)$ and the additional modes used by T_{new} all have $\lambda_j \leq \gamma(D)$. Then $S(T') = S(T)$, so by Step B $\mathcal{R}^*(T') = \mathcal{R}^*(T)$ and $d_{\text{eff}}(T') = d_{\text{eff}}(T)$. Coordinates outside $\sigma(\phi)$ cannot reduce risk by Step A. $\square$

Remark 1 (Empirical estimators). *Two estimators of d_{eff} appear in the main paper. The participation ratio $d_{\text{PR}} = (\sum_j \lambda_j)^2 / \sum_j \lambda_j^2$ is a global concentration measure and provides an upper bound on $r(\gamma)$ for moderate γ ; the Levina–Bickel estimator [6] estimates the local manifold dimension d_{loc} , which controls $\mu(C(z, \theta))$ in Theorem 2. Both are consistent with Theorem 1, which bounds d_{eff} from above by $r(\gamma(D))$ without pinning a single estimator.*

Remark 2 (Empirical $d_{\text{intrinsic}}$). *The range $d_{\text{intrinsic}} \approx 10$ –50 reported for natural language [6, 7] is an observed property of the spectrum of natural-language semantic kernels, not a mathematical consequence of the theorem. Theorem 1 says: given a kernel with a finite tail beyond threshold $\gamma(D)$, the effective rank is finite. It does not predict the numerical value of $r(\gamma(D))$ for any specific corpus.*

S3 Full proof of Theorem 2 (Positive Cap Mass)

Theorem (Positive Cap Mass; main paper, Theorem 2). *Under Axioms A1 and A4, for any anchor z and sufficiently small retrieval radius θ ,*

$$c'_1 \theta^{d_{\text{loc}}(z)} \leq \mu(C(z, \theta)) \leq c'_2 \theta^{d_{\text{loc}}(z)}.$$

Every admissible retrieval neighbourhood has strictly positive competitor mass.

Proof. By Axiom A1, items reside in the Hilbert space $\mathcal{H}_K$ via the canonical feature map ϕ . Without loss of generality, restrict to the unit sphere $\mathbb{S}_{\mathcal{H}} = \{u \in \mathcal{H}_K : \|u\| = 1\}$; the general case follows by homogeneity of the kernel score. The retrieval cap of radius θ at anchor z is

$$C(z, \theta) = \{x : \langle \phi(z), \phi(x) \rangle_{\mathcal{H}_K} \geq \cos \theta\}.$$

Equivalence of spherical cap and ball. For $u, v \in \mathbb{S}_{\mathcal{H}}$, $\|u - v\|^2 = 2 - 2\langle u, v \rangle$, so $\langle \phi(z), \phi(x) \rangle \geq \cos \theta$ is equivalent to $\|\phi(z) - \phi(x)\|^2 \leq 2(1 - \cos \theta) = 4 \sin^2(\theta/2)$. Hence

$$C(z, \theta) = \phi^{-1}(B(\phi(z), 2 \sin(\theta/2))),$$

where $B(\phi(z), r) = \{u \in \mathcal{H}_K : \|u - \phi(z)\| \leq r\}$ is the Hilbert-space ball.

Application of Axiom A4. For $\theta < \theta_0(z)$, $r = 2 \sin(\theta/2) < r_0(z)$ where $r_0(z)$ is the regularity radius. By Axiom A4 applied to the pushforward $\mu = \phi_{\#} P_X$,

$$c_1 r^{d_{\text{loc}}(z)} \leq \mu(B(\phi(z), r)) \leq c_2 r^{d_{\text{loc}}(z)}.$$

Substituting $r = 2 \sin(\theta/2)$ and using $\sin(\theta/2) = \theta/2 + O(\theta^3)$ for small θ ,

$$r^{d_{\text{loc}}(z)} = \theta^{d_{\text{loc}}(z)} (1 + O(\theta^2)).$$

Setting $c'_1 = c_1(1 - \epsilon_\theta)$ and $c'_2 = c_2(1 + \epsilon_\theta)$ with $\epsilon_\theta \rightarrow 0$ as $\theta \rightarrow 0$, we obtain, for $\theta < \theta_0(z)$,

$$c'_1 \theta^{d_{\text{loc}}(z)} \leq \mu(C(z, \theta)) \leq c'_2 \theta^{d_{\text{loc}}(z)}.$$

Strict positivity. Since $c'_1 > 0$ and $\theta > 0$, $\mu(C(z, \theta)) > 0$ for every admissible θ . Strict positivity persists for all $\theta \geq \theta_0(z)$ by monotonicity of measure. $\square$

Remark 3. The constants c'_1, c'_2 depend on z through $d_{\text{loc}}(z)$ and the regularity constants c_1, c_2 of Axiom A4. For natural-language semantic embeddings, d_{loc} is observed to lie in the range 10–50 [6]. For $\theta = 0.3$ (a typical retrieval radius) and $d_{\text{loc}} = 20$, $\theta^{d_{\text{loc}}} \approx 3.5 \times 10^{-11}$, small but strictly positive; a corpus of N items contains $\Theta(N \theta^{d_{\text{loc}}})$ competitors in expectation, which becomes order one for $N \gtrsim 10^{11}$. The relevant regime for production memory systems with $N = 10^6$ – 10^9 is $\mu(C) \in [10^{-5}, 10^{-2}]$, sufficient to drive the interference dynamics analysed in Theorem 3.

S4 Proofs of Theorem 3, Corollary 4, and Proposition 5

S4.1 Theorem 3 (Inevitable Forgetting Under Growing Memory)

Theorem (Inevitable Forgetting; main paper, Theorem 3). *Under Axioms A1 and A4, if competitor arrivals for item x form a marked point process with cumulative intensity $\Lambda_x(t) = \int_0^t \lambda_x(u) du$, retention satisfies*

$$R_x(t) = \exp(-\mu(C_x) \Lambda_x(t)).$$

If $\Lambda_x(t) \rightarrow \infty$, then $R_x(t) \rightarrow 0$.

Proof. Fix item x with retrieval cap $C_x = C(\phi(x), \theta)$ of radius θ . A retrieval competition event occurs at time u if and only if an arriving competitor lies in C_x . Under the marked Poisson model of Axiom A4 (competitors drawn i.i.d. from μ at rate $\lambda_x(u)$), the conditional probability that the u -th arrival falls in C_x is $\mu(C_x)$, independently of past arrivals.

Let $N_x(t)$ denote the number of competitors arriving in C_x during $[0, t]$. By the marking theorem [8, Ch. 5], $N_x(t)$ is itself a Poisson process with cumulative intensity

$$\mathbb{E}[N_x(t)] = \mu(C_x) \int_0^t \lambda_x(u) du = \mu(C_x) \Lambda_x(t).$$

Retention of x at time t is the probability that no competitor has yet occupied the cap, i.e. $R_x(t) = \mathbb{P}[N_x(t) = 0]$. For a Poisson random variable with mean $\mu(C_x)\Lambda_x(t)$, this probability is

$$R_x(t) = \exp(-\mu(C_x)\Lambda_x(t)).$$

By Theorem 2, $\mu(C_x) > 0$. If $\Lambda_x(t) \rightarrow \infty$ as $t \rightarrow \infty$, then $R_x(t) \rightarrow 0$. $\square$

Remark 4. *The result is robust to the precise retention rule: it requires only that retention is monotone non-increasing in the count of in-cap competitors and that retention drops to zero on the event of saturation. Variants in which retention degrades by a factor $\rho < 1$ per in-cap arrival yield $R_x(t) = \rho^{N_x(t)}$, with $\mathbb{E}[R_x(t)] = \exp(-(1-\rho)\mu(C_x)\Lambda_x(t))$, which has the same asymptotic behaviour.*

S4.2 Corollary 4 (Stretched Exponential Per-Item Retention)

Corollary (Stretched Exponential; main paper, Corollary 4). *If $\lambda_x(t) = \lambda_{0,x} t^{-\alpha}$ with $0 < \alpha < 1$, then*

$$R_x(t) = \exp(-c_x t^{1-\alpha}), \quad c_x = \frac{\mu(C_x) \lambda_{0,x}}{1-\alpha}.$$

Proof. Compute the cumulative intensity directly:

$$\Lambda_x(t) = \int_0^t \lambda_{0,x} u^{-\alpha} du = \lambda_{0,x} \frac{t^{1-\alpha}}{1-\alpha},$$

valid for $0 < \alpha < 1$ (which makes the integral converge at the lower limit when interpreted as $\lim_{\epsilon \rightarrow 0^+} \int_{\epsilon}^t$). Substituting into Theorem 3,

$$R_x(t) = \exp(-\mu(C_x)\Lambda_x(t)) = \exp\left(-\frac{\mu(C_x) \lambda_{0,x}}{1-\alpha} t^{1-\alpha}\right).$$

With $c_x = \mu(C_x) \lambda_{0,x}/(1-\alpha)$, this is precisely a Kohlrausch–Williams–Watts stretched exponential [9] of stretching index $1-\alpha$. $\square$

Remark 5. *A per-item curve $R_x(t) = \exp(-c_x t^{1-\alpha})$ is not a power law; in log-log coordinates it has curvature, not a constant slope. The population-level power-law forgetting curve observed empirically arises only after averaging over heterogeneity in c_x (Proposition 5 below). Treating an Ebbinghaus curve as power-law for individual items is therefore a category error; the power law is a population property.*

S4.3 Proposition 5 (Population Power Law from Heterogeneity)

Proposition (Population Power Law; main paper, Proposition 5). *If the item-specific scale c_x has a density regularly varying at zero, $g(c) \sim \kappa c^{\beta-1}$ as $c \downarrow 0$, then the population-averaged retention obeys*

$$\bar{R}(t) \sim \kappa \Gamma(\beta) t^{-\beta(1-\alpha)}.$$

The population forgetting exponent is $b = \beta(1-\alpha)$.

Proof. Population-averaged retention is

$$\bar{R}(t) = \int_0^\infty e^{-c t^{1-\alpha}} g(c) dc.$$

This is a Laplace transform $\mathcal{L}\{g\}(s)$ evaluated at $s = t^{1-\alpha}$:

$$\bar{R}(t) = \mathcal{L}\{g\}(t^{1-\alpha}).$$

By the Tauberian theorem for Laplace transforms of regularly varying densities at zero [10, Theorem XIII.5.4], if $g(c) \sim \kappa c^{\beta-1}$ as $c \rightarrow 0^+$ with $\beta > 0$, then

$$\mathcal{L}\{g\}(s) \sim \kappa \Gamma(\beta) s^{-\beta} \quad \text{as } s \rightarrow \infty.$$

Substituting $s = t^{1-\alpha}$,

$$\bar{R}(t) \sim \kappa \Gamma(\beta) t^{-\beta(1-\alpha)} \quad \text{as } t \rightarrow \infty,$$

which is a power-law decay with exponent $b = \beta(1 - \alpha)$. $\square$

Remark 6 (Quantitative range). *The exponent α is corpus-dependent. Anderson & Schooler reported $\alpha = 0.513$ on newspaper text [11]; the main paper reports $\alpha = 0.459$ on Wikipedia. With $\beta \in [0.6, 1.4]$, both place $b = \beta(1 - \alpha)$ in the empirically observed $[0.3, 0.7]$ range. The geometric layer (Theorem 2) controls $\mu(C_x)$ and therefore the constant κ ; the temporal layer (Anderson–Schooler α) and population heterogeneity (β) jointly control the exponent.*

S5 Proof of Theorem 7 (No Escape for Kernel-Threshold Memory)

Theorem (No Escape; main paper, Theorem 7). *Under Axioms A1–A5: (1) the semantically useful representation has effective rank controlled by the semantic operator spectrum; (2) every admissible retrieval neighbourhood has positive competitor mass; (3) under growing memory, retention decays to zero; (4) for δ -convex associative lures with δ below the decision margin, false recall cannot be eliminated by threshold tuning within the same score family. Any architecture that simultaneously eliminates interference-driven forgetting and associative false recall must either abandon semantic continuity and kernel-threshold retrieval, add an external symbolic verifier or exact episodic record, or send the semantic effective rank to infinity.*

Proof. Claims (1)–(4) are restatements of Theorems 1, 2, 3, and 6 of the main paper, established in §S2, §S3, §S4, and the inline proof of Theorem 6 in the main paper, respectively. It remains to prove the negative claim: that no kernel-threshold architecture can simultaneously eliminate both interference-driven forgetting and associative false recall, except by violating one of A1–A5. Suppose, for contradiction, that some encoder T and threshold τ satisfy:

- (a) No interference-driven forgetting: $R_x(t) \rightarrow R_\infty(x) > 0$ for all x as the corpus grows;
- (b) No associative false recall: for every δ -convex lure c with $\delta < m$ (margin), $f_q(c) < \tau$.

Consequence of (a). By Theorem 3, $R_x(t) = \exp(-\mu(C_x)\Lambda_x(t))$. If competitor arrivals are non-trivial ($\Lambda_x(t) \rightarrow \infty$), $R_x(t) \rightarrow 0$ unless $\mu(C_x) = 0$. By Theorem 2, $\mu(C_x) = 0$ requires the cap C_x to have zero pushforward measure, which under Axiom A4 (local Ahlfors regularity with $d_{\text{loc}}(z) < \infty$) is impossible for any positive radius. Hence (a) requires one of the following:

- (a1) $\Lambda_x(t) \not\rightarrow \infty$, i.e. the corpus does not grow without bound. Excluded by the “growing memory” premise.
- (a2) Axiom A4 fails, i.e. $d_{\text{loc}}(z) = \infty$ on a set of positive measure. By Theorem 1, this requires the semantic effective rank to be infinite, which contradicts rate-distortion optimality (Axiom A3) unless the rate or distortion budget is infinite.
- (a3) Axiom A1 fails, i.e. the score is not a monotone function of an inner-product in $\mathcal{H}_K$. This is the SPP-violating route taken by BM25 in Architecture 3.
- (a4) An external symbolic verifier is appended: an oracle that, given a candidate retrieval, returns ground-truth relevance without using the kernel score. This circumvents the cap-mass argument by replacing the score-based decision rule.

Consequence of (b). By Theorem 6 (inline proof in the main paper), for any δ -convex lure c with $\delta < m$, $f_q(c) \geq \tau + m - \delta > \tau$, so $f_q(c) > \tau$ and the lure is accepted. To make $f_q(c) < \tau$ for all such lures, one of the following must hold:

- (b1) Axiom A5 fails on the lure: no convex combination of studied items lies within δ of the lure in $\mathcal{H}_K$. This restricts to non-associative recognition tasks; for DRM-style lures the empirical $\delta^* < m$ (main paper, Theorem 6 discussion).
- (b2) Axiom A1 fails, as in (a3).
- (b3) An external verifier is appended, as in (a4): the lure is rejected by a non-score check, e.g. exact list match in attention memory.

Combining. To satisfy (a) and (b) simultaneously, the encoder must invoke at least one of the escape routes (a2), (a3), (a4), (b1), (b2), or (b3). Each escape route corresponds exactly to one of the three options stated in the theorem:

- Routes (a3) and (b2) abandon Axiom A1: semantic continuity and kernel-threshold retrieval are dropped. This is the “abandon SPP” option.
- Routes (a4) and (b3) add an external symbolic verifier or exact episodic record. This is the “add external verifier” option.
- Route (a2) sends the semantic effective rank to infinity (by Theorem 1, this requires either an infinite rate budget or violation of Axiom A3). This is the “send effective rank to infinity” option.

Route (b1) is excluded by the empirical premise of the theorem ($\delta^* < m$ for all DRM lures tested). No other route exists within the kernel-threshold class defined by Axioms A1–A5. $\square$

Remark 7 (Empirical correspondence). *The three escape routes manifest in the five-architecture comparison of the main paper as follows. The filesystem (Architecture 3, BM25) takes the “abandon Axiom A1” route: it eliminates both forgetting and false recall but at the cost of 15.5% semantic agreement with cosine-based retrieval. Attention and parametric architectures (2, 5) take the “add external verifier” route for DRM lures (FA = 0.000 behaviourally) while remaining geometrically vulnerable (lures still within caps; phase-transition failure under growing memory). No tested architecture takes the “infinite rank” route, because doing so would violate Axiom A3 (rate-distortion optimality); the closest analogue—zero-padding to higher nominal dimension—leaves d_{eff} and b unchanged, exactly as Theorem 1 predicts.*

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